

ELKO, NV, USA

WMO: 725825

Lat: 40.829N

Lon: 115.789W

Elev: 1533

StdP: 84.21

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 24121

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.4	-15.3	-21.4	0.7	-16.4	-18.7	0.9	-11.7	10.6	4.3	8.7	3.7	1.5	80	0.596

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.0	35.3	15.1	33.9	14.6	32.5	14.1	16.7	30.4	15.9	29.5	15.2	28.8	3.9	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.5	10.9	19.4	10.8	9.7	19.6	9.3	8.8	19.5	52.0	30.2	49.6	29.1	47.3	28.6	21.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.8	8.3	7.3	-22.5	37.4	3.6	1.7	-25.1	38.6	-27.3	39.6	-29.3	40.5	-32.0	41.7		
			-22.9	18.2	3.5	1.2	-25.4	19.0	-27.5	19.7	-29.5	20.4	-32.1	21.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.6	-2.9	-0.8	4.1	7.3	12.0	17.3	22.2	20.8	15.5	8.3	2.2	-2.8
	DBStd	9.58	5.42	4.87	4.02	3.93	4.15	4.05	2.87	2.74	4.08	4.29	4.85	5.07
	HDD10.0	1741	401	302	186	98	27	3	0	0	8	83	236	398
	HDD18.3	3797	659	535	442	331	198	64	4	10	101	312	485	656
	CDD10.0	1244	0	0	3	17	89	222	378	333	171	29	1	0
	CDD18.3	259	0	0	0	0	3	32	124	85	14	1	0	0
	CDH23.3	6079	0	0	1	15	181	988	2369	1860	618	48	0	0
	CDH26.7	2911	0	0	0	2	41	418	1290	941	213	7	0	0
Wind	WSAvg	2.5	2.1	2.4	2.7	3.0	2.9	2.8	2.6	2.4	2.2	2.1	2.1	2.1
Precipitation	PrecAvg	252	28	21	24	22	27	20	9	14	15	18	27	28
	PrecMax	466	83	59	61	55	104	66	37	117	82	50	70	107
	PrecMin	121	1	2	1	0	0	0	0	0	0	0	2	0
	PrecStd	77	20	13	16	14	22	17	9	21	18	15	15	25
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.3	15.7	21.2	25.3	30.2	35.5	37.6	35.8	32.7	27.5	19.5	11.9
		MCWB	4.8	6.2	8.3	10.7	13.2	15.2	16.4	15.0	14.1	11.9	8.5	5.2
	2%	DB	8.8	12.1	18.3	22.5	27.5	32.9	36.1	34.4	30.9	24.3	16.5	8.8
		MCWB	3.6	5.2	7.3	9.4	12.4	13.9	15.5	14.5	13.1	10.1	7.1	4.0
	5%	DB	6.8	9.1	15.9	19.8	25.4	31.1	34.7	33.2	29.2	21.9	13.6	6.9
		MCWB	2.7	3.7	6.4	8.3	11.2	13.4	15.0	14.1	12.5	9.0	5.7	2.7
	10%	DB	4.5	6.7	13.1	17.1	22.7	29.1	33.1	32.0	27.4	19.1	10.9	4.7
		MCWB	1.4	2.5	5.3	7.3	10.4	12.9	14.6	13.8	12.1	8.1	4.7	1.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.3	7.1	9.2	11.6	14.6	16.3	18.0	17.1	16.2	12.6	9.4	6.3
		MCDB	9.4	13.8	19.3	23.8	27.6	32.2	33.0	29.7	26.7	25.3	17.0	10.1
	2%	WB	4.3	5.5	7.8	9.9	13.0	15.1	17.0	16.2	14.7	11.1	7.6	4.7
		MCDB	7.8	11.1	17.1	20.5	24.8	29.9	31.2	29.2	26.1	21.8	15.1	8.2
	5%	WB	2.9	4.1	6.7	8.8	11.9	14.1	16.3	15.5	13.7	9.8	6.3	3.1
		MCDB	5.9	8.4	15.1	18.4	23.0	28.7	30.3	28.8	26.2	19.4	12.9	6.2
	10%	WB	1.6	2.8	5.6	7.7	10.9	13.4	15.6	14.7	12.6	8.7	5.0	1.8
		MCDB	4.3	6.6	12.5	16.3	21.4	27.4	29.4	28.5	25.5	18.1	10.1	4.4
Mean Daily Temperature Range	5% DB	MDBR	11.6	11.9	14.3	14.7	16.4	19.3	21.0	21.4	20.7	18.9	14.8	11.6
		MCDBR	14.0	14.7	19.4	21.1	21.4	23.0	22.6	23.1	23.5	23.7	19.3	13.4
		MCWBR	9.5	8.9	10.1	10.1	9.0	8.8	8.3	9.1	10.4	12.2	11.3	9.1
	5% WB	MCDBR	12.1	13.4	17.9	18.7	19.0	21.3	19.6	20.0	20.3	20.7	17.8	11.2
	MCWBR	8.4	8.4	9.6	9.1	8.2	8.5	7.4	8.1	9.0	10.8	10.5	7.8	
Clear-Sky Solar Irradiance	taub	0.244	0.258	0.271	0.299	0.309	0.311	0.330	0.320	0.300	0.271	0.255	0.245	
	taud	2.528	2.552	2.562	2.476	2.495	2.518	2.482	2.486	2.542	2.614	2.600	2.555	
	Ebn at Noon	944	976	993	975	966	960	938	941	945	948	924	916	
	Edh at Noon	70	80	90	105	106	104	107	103	91	75	65	63	
All-Sky Solar Radiation	RadAvg	2.02	2.92	4.18	5.45	6.55	7.80	7.52	6.72	5.54	3.85	2.43	1.65	
	RadStd	0.27	0.37	0.32	0.38	0.67	0.53	0.42	0.39	0.29	0.26	0.18	0.28	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.60	N/A	N/A	+1.06	N/A	N/A	N/A	-155	+203	+109
(47)	Regional (5 neighbors)	N/A	N/A	N/A	+0.61	N/A	N/A	N/A	N/A	+121	+57

Nomenclature: See separate page